EXPERIMENT-1:

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Code:

```
# ==========================================

# FIND-S Algorithm

# ==========================================


import pandas as pd

from google.colab import files

# Upload CSV File

uploaded = files.upload()

# Read Dataset

filename = list(uploaded.keys())[0]

data = pd.read_csv(filename)

print("Training Dataset:\n")

print(data)

# Separate features and target

concepts = data.iloc[:, :-1].values

target = data.iloc[:, -1].values

# Initialize hypothesis

hypothesis = ['0'] * len(concepts[0])

print("\nInitial Hypothesis:")

print(hypothesis)

# FIND-S Algorithm

for i in range(len(concepts)):

    if target[i].strip().lower() == "yes":

        for j in range(len(hypothesis)):

            if hypothesis[j] == '0':

                hypothesis[j] = concepts[i][j]
```

```
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
        print(f"\nStep {i+1}:")
        print("Hypothesis:", hypothesis)
print("\n=================================")
print("Final Most Specific Hypothesis:")
print(hypothesis)
print("=================================")
```

Output:

```
Choose Files  EnjoySport(1).csv
EnjoySport(1).csv(text/csv) - 435 bytes, last modified: 8/3/2026 - 100% done
Saving EnjoySport(1).csv to EnjoySport(1) (1).csv
Training Dataset:

         Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0      Sunny    Warm   Normal  Strong  Warm     Same        Yes
1      Sunny    Warm     High  Strong  Warm     Same        Yes
2      Rainy    Cold     High  Strong  Warm   Change         No
3      Sunny    Warm     High  Strong  Cool   Change        Yes
4      Sunny    Cold   Normal    Weak  Warm     Same        Yes
5      Rainy    Warm   Normal    Weak  Cool   Change         No
6   Overcast    Warm     High  Strong  Warm     Same        Yes
7   Overcast    Cool   Normal    Weak  Cool     Same        Yes
8      Rainy    Cool     High  Strong  Cool   Change         No
9      Sunny    Warm   Normal    Weak  Warm     Same        Yes

Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Step 1:
Hypothesis: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Step 2:
Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Step 3:
Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Step 4:
Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

```
Step 4:
Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

Step 5:
Hypothesis: ['Sunny', '?', '?', '?', '?', '?']

Step 6:
Hypothesis: ['Sunny', '?', '?', '?', '?', '?']

Step 7:
Hypothesis: ['?', '?', '?', '?', '?', '?']

Step 8:
Hypothesis: ['?', '?', '?', '?', '?', '?']

Step 9:
Hypothesis: ['?', '?', '?', '?', '?', '?']

Step 10:
Hypothesis: ['?', '?', '?', '?', '?', '?']

=================================
Final Most Specific Hypothesis:
['?', '?', '?', '?', '?', '?']
=================================
```